Characteristics of pollen-related food allergy based on individual pollen allergy profiles in the Chinese population.

PMID: 32435327 | Indexed in PubMed

Patients with pollinosis are often multi-sensitized to diverse pollen allergens. However, little is known about pollen allergy types among Chinese pollinosis patients. This study is aimed to characterize clinical manifestations of food allergy among patients with different types of pollen allergy. We retrospectively analyzed 402 pollinosis patients from an outpatient allergy department of the Peking Union Medical College Hospital who had been diagnosed by experienced allergists. All included patients who answered a questionnaire regarding seasonal pollinosis and clinical symptoms after ingestion of food and underwent intradermal skin tests. Total IgE and specific IgE levels were quantified by ImmunoCAP, using 0.35 kUA/L as a threshold for positivity. The patients were divided into 3 groups, based on the season during which they experienced symptoms and the 2 peaks of Chinese airborne pollen: spring-tree pollen symptoms group (SG), autumn-weed pollen symptoms group (AG), and a combined spring and autumn pollen symptoms group (CG). Birch pollen (83%) and ash pollen (74%) were common allergens among patients with spring symptoms, while mugwort pollen (87%) was a common allergen among patients with autumn symptoms. In total, 30% of the study population experienced pollen-related food allergy. Pollen-related food allergies were more prevalent among the single-season symptom groups (68% and 50% for the SG and AG, respectively) than among the CG (20%). All patients with pollen-related food-induced anaphylaxis exhibited autumn weed pollen symptoms. Except for 2 patients, all patients with food-induced anaphylaxis were allergic to mugwort pollen. In the SG, all patients with food allergy were sensitive to birch pollen, with birch and Bet v 1-specific IgE levels higher in this group than in the group without food allergy ($p < 0.001$). In the AG, Art v 3 was more prevalent among patients with pollen-related food allergy than without food allergy (79% vs. 33%, $p < 0.001$), a proportion similar to the one in the CG (67% vs. 48%, $p = 0.01$). Meanwhile, the Art v 3-specific IgE levels among patients with pollen-related food allergy were higher than among tolerant patients in the AG ($p < 0.001$) and CG ($p = 0.02$). Unexpectedly, the Art v 3-specific IgE levels were higher in patients with food-induced anaphylaxis than with oral allergy syndrome only in the CG. Bet v 1 (a Pathogenesis-related 10 protein) and Art v 3 (a non-specific lipid transfer protein; nsLTP) are candidate molecular biomarkers for the diagnosis of food allergies in patients with season-specific pollen-related allergies. Measuring pollen allergen component-specific IgE levels might be an effective tool for the management of pollinosis in clinical practice in China.

Pollen-related allergy in Europe.

PMID: 9689338 | Indexed in PubMed

The increasing mobility of Europeans for business and leisure has led to a need for reliable information about exposure to seasonal airborne allergens during travel abroad. Over the last 10 years or so, aeropalynologic and allergologic studies have progressed to meet this need, and extensive international networks now provide regular pollen and hay-fever forecasts. Europe is a geographically complex continent with a widely diverse climate and a wide spectrum of vegetation. Consequently, pollen calendars differ from one area to another; however, on the whole, pollination starts in spring and ends in autumn. Grass pollen is by far the most frequent cause of pollinosis in Europe. In northern Europe, pollen from species of the family Betulaceae is a major cause of the disorder. In contrast, the mild winters and dry summers of Mediterranean areas favor the production of pollen types that are rarely found in central and northern areas of the continent (e.g., the genera Parietaria, Olea, and Cupressus). Clinical and aerobiologic studies show that the pollen map of Europe is changing also as a result of cultural factors (e.g., importation of plants for urban parklands) and greater international travel (e.g., the expansion of the ragweed genus Ambrosia in France, northern Italy, Austria, and Hungary). Studies on allergen-carrying paucimicronic or submicronic airborne particles, which penetrate deep into the lung, are having a relevant impact on our understanding of pollinosis and its distribution throughout Europe.

IgE cross-reactivity between Lolium multiflorum and commercial grass pollen allergen extracts in Brazilian patients with pollinosis.

PMID: 20098843 | Indexed in PubMed

Lolium multiflorum (Lm) grass pollen is the major cause of pollinosis in Southern Brazil. The objectives of this study were to investigate immunodominant components of Lm pollen allergens and the cross-reactivity of IgE with commercial grass pollen allergen extracts. Thirty-eight serum samples from patients with seasonal allergic rhinitis (SAR), 35 serum samples from patients with perennial allergic rhinitis (PAR) and 30 serum samples from non-atopic subjects were analyzed. Allergen sensitization was evaluated using skin prick test and serum IgE levels against Lm pollen extract were determined by ELISA. Inhibition ELISA and immunoblot were used to evaluate the cross-reactivity of IgE between allergens from Lm and commercial grass pollen extracts, including L. perenne (Lp), grass mix I (GI) and II (GII) extracts. IgE antibodies against Lm were detected in 100% of SAR patients and 8.6% of PAR patients. Inhibition ELISA demonstrated IgE cross-reactivity between homologous (Lm) and heterologous (Lp or GII) grass pollen extracts, but not for the GI extract. Fifteen IgE-binding Lm components were detected and immunoblot bands of 26, 28-30, and 32-35 kDa showed >90% recognition. Lm, Lp and GII extracts significantly inhibited IgE binding to the most immunodominant Lm components, particularly the 55 kDa band. The 26 kDa and 90-114 kDa bands presented the lowest amount of heterologous inhibition. We demonstrated that Lm extract contains both Lm-specific and cross-reactive IgE-binding components and therefore it is suitable for measuring quantitative IgE levels for diagnostic and therapeutic purposes in patients with pollinosis sensitized to Lm grass pollen rather than other phylogenetically related grass pollen extracts.

The 3D enhancer network of the developing T cell genome is shaped by SATB1.

PMID: 36376298 | Indexed in PubMed

Mechanisms of tissue-specific gene expression regulation via 3D genome organization are poorly understood. Here we uncover the regulatory chromatin network of developing T cells and identify SATB1, a tissue-specific genome organizer, enriched at the anchors of promoter-enhancer loops. We have generated a T-cell specific Satb1 conditional knockout mouse which allows us to infer the molecular mechanisms responsible for the deregulation of its immune system. H3K27ac HiChIP and Hi-C experiments indicate that SATB1-dependent promoter-enhancer loops regulate expression of master regulator genes (such as Bcl6), the T cell receptor locus and adhesion molecule genes, collectively being critical for cell lineage specification and immune system homeostasis. SATB1-dependent regulatory chromatin loops represent a more refined layer of genome organization built upon a high-order scaffold provided by CTCF and other factors. Overall, our findings unravel the function of a tissue-specific factor that controls transcription programs, via spatial chromatin arrangements complementary to the

chromatin structure imposed by ubiquitously expressed genome organizers.

Vascular Immune Evasion of Mesenchymal Glioblastoma Is Mediated by Interaction and Regulation of VE-Cadherin on PD-L1.

PMID: 37686533 | Indexed in PubMed

The mesenchymal subtype of glioblastoma (mGBM), which is characterized by rigorous vasculature, resists anti-tumor immune therapy. Here, we investigated the mechanistic link between tumor vascularization and the evasion of immune surveillance. Clinical datasets with GBM transcripts showed that the expression of the mesenchymal markers YKL-40 (CHI3L1) and Vimentin is correlated with elevated expression of PD-L1 and poor disease survival. Interestingly, the expression of PD-L1 was predominantly found in vascular endothelial cells. Orthotopic transplantation of glioma cells GL261 over-expressing YKL-40 in mice showed increased angiogenesis and decreased CD8+ T cell infiltration, resulting in a reduction in mouse survival. The exposure of recombinant YKL-40 protein induced PD-L1 and VE-cadherin (VE-cad) expression in endothelial cells and drove VE-cad-mediated nuclear translocation of β-catenin/LEF, where LEF upregulated PD-L1 expression. YKL-40 stimulated the dissociation of VE-cad from PD-L1, rendering PD-L1 available to interact with PD-1 from CD8+-positive TALL-104 lymphocytes and inhibit TALL-104 cytotoxicity. YKL-40 promoted TALL-104 cell migration and adhesion to endothelial cells via CCR5-dependent chemotaxis but blocked its anti-vascular immunity. Knockdown of VE-cad or the PD-L1 gene ablated the effects of YKL-40 and reinvigorated TALL-104 cell immunity against vessels. In summary, our study demonstrates a novel vascular immune escape mechanism by which mGBM promotes tumor vascularization and malignant transformation.

Immunotherapy drives mesenchymal tumor cell state shift and TME immune response in glioblastoma patients.

PMID: 38695342 | Indexed in PubMed

Glioblastoma is a highly aggressive type of brain tumor for which there is no curative treatment available. Immunotherapies have shown limited responses in unselected patients, and there is an urgent need to identify mechanisms of treatment resistance to design novel therapy strategies. Here we investigated the phenotypic and transcriptional dynamics at single-cell resolution during nivolumab immune checkpoint treatment of glioblastoma patients. We present the integrative paired single-cell RNA-seq analysis of 76 tumor samples from patients in a clinical trial of the PD-1 inhibitor nivolumab and untreated patients. We identify a distinct aggressive phenotypic signature in both tumor cells and the tumor microenvironment in response to nivolumab. Moreover, nivolumab-treatment was associated with an increased transition to mesenchymal stem-like tumor cells, and an increase in TAMs and exhausted and proliferative T cells. We verify and extend our findings in large external glioblastoma dataset (n = 298), develop a latent immune signature and find 18% of primary glioblastoma samples to be latent immune, associated with mesenchymal tumor cell state and TME immune response. Finally, we show that latent immune glioblastoma patients are associated with shorter overall survival following immune checkpoint treatment (P = .0041). We find a resistance mechanism signature in one fifth of glioblastoma patients associated with a tumor-cell transition to a more aggressive mesenchymal-like state, increase in TAMs and proliferative and exhausted T cells in response to immunotherapy. These patients may instead

benefit from neuro-oncology therapies targeting mesenchymal tumor cells.

Elucidating the diversity of malignant mesenchymal states in glioblastoma by integrative analysis.

PMID: 36123598 | Indexed in PubMed

Multiple glioblastoma studies have described a mesenchymal (MES) state, with each study defining the MES program by distinct sets of genes and highlighting distinct functional associations, including both immune activation and suppression. These variable descriptions complicate our understanding of the MES state and its implications. Here, we hypothesize that there is a range of glioma MES states, possibly reflecting distinct prior states in which a MES program can be induced, and/or distinct mechanisms that induce the MES states in those cells. We integrated multiple published single-cell and bulk RNA sequencing datasets and MES signatures to define a core MES program that recurs across studies, as well as multiple function-specific MES signatures that vary across MES cells. We then examined the co-occurrence of these signatures and their associations with genetic and microenvironmental features. Based on co-occurrence of MES signatures, we found three main variants of MES states: hypoxia-related (MES-Hyp), astrocyte-related (MES-Ast), and an intermediate state. Notably, the MES states are differentially associated with genetic and microenvironmental features. MES-Hyp is preferentially associated with NF1 deletion, overall macrophage abundance, a high macrophage/microglia ratio, and M2-related macrophages, consistent with previous studies that associated MES with immune suppression. In contrast, MES-Ast is associated with T cell abundance and cytotoxicity, consistent with immune activation through expression of MHC-I/II. Diverse MES states occur in glioblastoma. These states share a subset of core genes but differ primarily in their association with hypoxia vs. astrocytic expression programs, and with immune suppression vs. activation, respectively.

Interactions between cancer cells and immune cells drive transitions to mesenchymal-like states in glioblastoma.

PMID: 34087162 | Indexed in PubMed

The mesenchymal subtype of glioblastoma is thought to be determined by both cancer cell-intrinsic alterations and extrinsic cellular interactions, but remains poorly understood. Here, we dissect glioblastoma-to-microenvironment interactions by single-cell RNA sequencing analysis of human tumors and model systems, combined with functional experiments. We demonstrate that macrophages induce a transition of glioblastoma cells into mesenchymal-like (MES-like) states. This effect is mediated, both in vitro and in vivo, by macrophage-derived oncostatin M (OSM) that interacts with its receptors (OSMR or LIFR) in complex with GP130 on glioblastoma cells and activates STAT3. We show that MES-like glioblastoma states are also associated with increased expression of a mesenchymal program in macrophages and with increased cytotoxicity of T cells, highlighting extensive alterations of the immune microenvironment with potential therapeutic implications.

Human Mesenchymal glioblastomas are characterized by an increased immune cell

presence compared to Proneural and Classical tumors.

PMID: 31646100 | Indexed in PubMed

Glioblastoma (GBM) is the most aggressive malignant primary brain tumor in adults, with a median survival of 14.6 months. Recent efforts have focused on identifying clinically relevant subgroups to improve our understanding of pathogenetic mechanisms and patient stratification. Concurrently, the role of immune cells in the tumor microenvironment has received increasing attention, especially T cells and tumor-associated macrophages (TAM). The latter are a mixed population of activated brain-resident microglia and infiltrating monocytes/monocyte-derived macrophages, both of which express ionized calcium-binding adapter molecule 1 (IBA1). This study investigated differences in immune cell subpopulations among distinct transcriptional subtypes of GBM. Human GBM samples were molecularly characterized and assigned to Proneural, Mesenchymal or Classical subtypes as defined by NanoString nCounter Technology. Subsequently, we performed and analyzed automated immunohistochemical stainings for TAM as well as specific T cell populations. The Mesenchymal subtype of GBM showed the highest presence of TAM, CD8+, CD3+ and FOXP3+ T cells, as compared to Proneural and Classical subtypes. High expression levels of the TAM-related gene AIF1, which encodes the TAM-specific protein IBA1, correlated with a worse prognosis in Proneural GBM, but conferred a survival benefit in Mesenchymal tumors. We used our data to construct a mathematical model that could reliably identify Mesenchymal GBM with high sensitivity using a combination of the aforementioned cell-specific IHC markers. In conclusion, we demonstrated that molecularly distinct GBM subtypes are characterized by profound differences in the composition of their immune microenvironment, which could potentially help to identify tumors amenable to immunotherapy.

Macrophage-tumor cell intertwine drives the transition into a mesenchymal-like cellular state of glioblastoma.

PMID: 34087163 | Indexed in PubMed

Macrophages are the major non-neoplastic infiltrates in the glioblastoma microenvironment. In this issue of Cancer Cell, Hara et al. (2021) demonstrate that macrophages induce a transition of glioblastoma cells into the mesenchymal-like cellular state associated with an increased mesenchymal program in macrophages themselves and enhanced cytotoxicity of T cells.